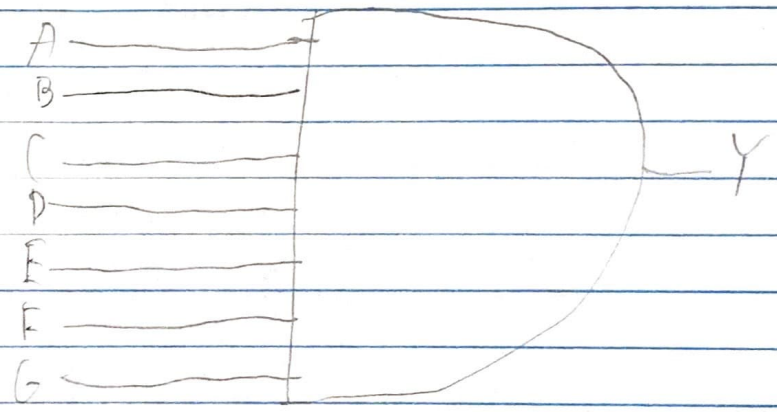
Dylan Thornton
4-9-23

1. a 001100 = 8+4 = 12
b 111100 = 16+8+4 = 28
c 000111100 = 32+16+8+4 = 60
d 101010 = 32+8+2 = 42

I used this to convert and just added the ones.

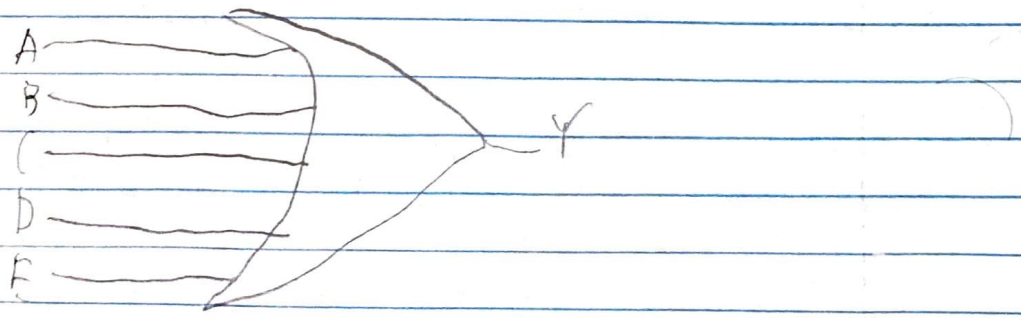
1 1 1 1 1 1 1 1
2^8 2^7 2^6 2^5 2^4 2^2 2^1 2^0

2. a.



ABCDEFG = Y
A∧B∧C∧D∧E∧F∧G = Y

b.

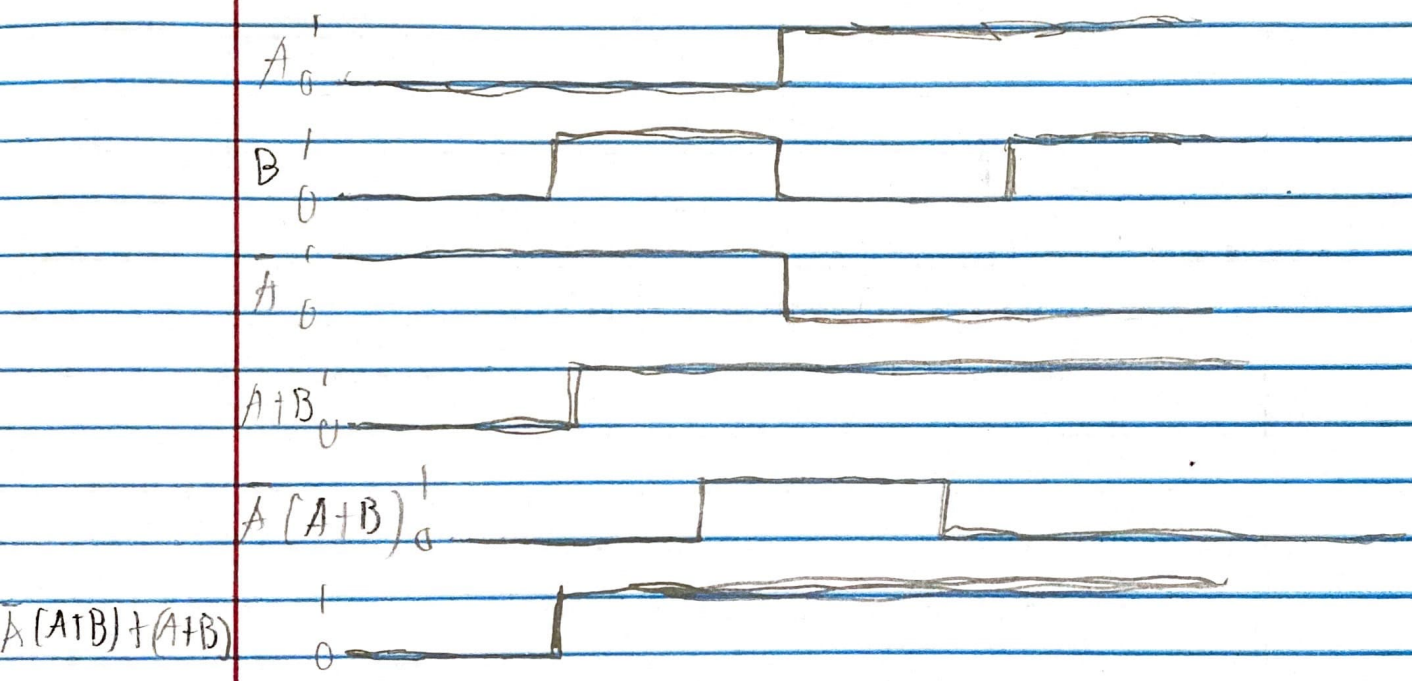


A+B+C+D+E = Y
A∨B∨C∨D∨E = Y

3. Circuit 1:

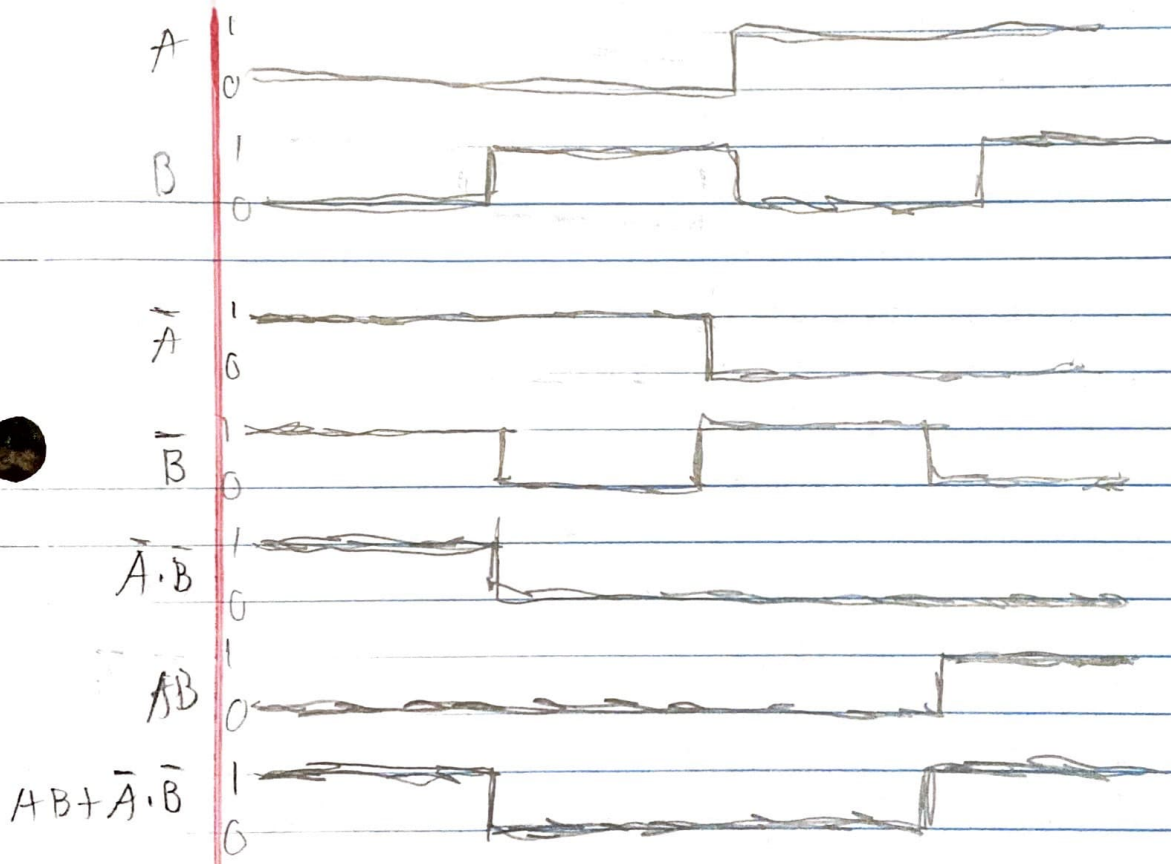
2

A	B	$\bar{A}$	$A+B$	$\bar{A}(A+B)$	$\bar{A}(A+B)+(A+B)$
0	0	1	0	0	0
0	1	1	1	1	1
1	0	0	1	0	1
1	1	0	1	0	1



Firmit 2:

A	B	$\bar{A}$	$\bar{B}$	AB	$\bar{A}\cdot\bar{B}$	$\overbrace{AB+\bar{A}\cdot\bar{B}}^{X}$
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1



4a,

C	B	A	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

$Y = \bar{C}\bar{B}\bar{A} + \bar{C}\bar{B}A + \bar{C}B\bar{A}$

$Y = \bar{C}(\bar{B}\bar{A} + \bar{B}A + B\bar{A})$

$Y = \bar{C}(\bar{B} + B\bar{A})$

$\boxed{Y = \bar{C}(\bar{B} + \bar{A})}$

A \ CB	00	01	11	10
0	1	1	0	0
1	1	0	0	0

$\bar{C}\bar{A} + \bar{C}\bar{B} = Y$

4b.

5c.

D	C	B	A	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

$Y = \bar{D}\bar{C}\bar{B}\bar{A} + \bar{D}\bar{C}B\bar{A} + \bar{D}C\bar{B}\bar{A} + \bar{D}CB\bar{A} + DCBA$

$Y = \bar{D}\bar{C}\bar{A}(\bar{B}+B) + D(\bar{C}\bar{B}\bar{A} + \bar{C}B\bar{A} + CBA)$

$Y = \bar{D}\bar{C}\bar{A} + D(\bar{C}\bar{A} + CBA)$

$Y = \bar{D}\bar{C}\bar{A} + D\bar{C}\bar{A} + ABCD$

$\boxed{Y = \bar{C}\bar{A} + ABCD}$

5b.

BA \ DC	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	1	0
10	1	0	0	1

$\bar{D}\bar{C}\bar{A} + D\bar{C}\bar{A} + DCBA$

$\bar{C}\bar{A}(\bar{D}+D) + ABCD$

$\boxed{\bar{C}\bar{A} + ABCD = Y}$